

# L6 K3 Gate 9 Cal #29 Question-Shape Audit RIGOROUS Extension

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## L6 K3 Gate 9 Cal #29 Question-Shape Audit RIGOROUS v0.1

### 0. Goal

Per Casey Thursday PM L6 + K3 v0.16 Gate 9: Cal #29 question-shape audit RIGOROUS extension covering Thursday substrate-mechanism candidates Composite v0.4 + IP-5 W mass + Cat 6/2/3/5 cascade work.

### 1. Cal #29 Question-Shape Audit Discipline

Per Cal #29 STANDING question-shape audit + Cal #35 STANDING form-selection  $\neq$  mechanism-forcing:

For each substrate-mechanism candidate, audit: - (a) Was form FORWARD-derived from substrate first-principles? - (b) Or was form SELECTED post-hoc from observed value matching?

substantive RIGOROUS-tier requires (a) forward-derivation + Tier 1 EXACT precision + Cal #35 STANDING form-selection brake-cleared.

### 2. Thursday Afternoon Candidates Question-Shape Audit

| Candidate                                                                                                  | Form                                        | Cal #29 audit                                                                                                                  | Cal #35 audit                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cat 6<br>substrate-symplectic 4<br>candidates ( $\text{Pf}(\omega)$ , $\omega^2/2!$ ,<br>Tr ratio, $\mu$ ) | substrate-canonical at<br>V_fund            | Cal #36 positive-search<br>FORWARD substantive;<br>all walked back at<br>substrate-canonical level                             | Mode 5 form-fitting<br>REJECTED via Cal #35<br>STANDING brakes                                                                                        |
| Cat 5 substrate-multipole<br>$1/(2k+1)$ at $\text{Sym}^3$                                                  | "= 1/g" candidate                           | substantive Cal #36<br>positive-search<br>FORWARD; walked back<br>~3 min                                                       | integer-INDEX<br>coincidence per Cal #35<br>STANDING                                                                                                  |
| Composite v0.4 Term 1<br>( $n_C \cdot (5/2)_3$ )                                                           | substrate-Bergman +<br>substrate-Pochhammer | substantive<br>FORWARD-derived from<br>substrate-Bergman<br>kernel exponent<br>$n_C/\text{rank}$ + Schur II<br>Pochhammer rule | substantive<br>substrate-mechanism<br>FORCING candidate per<br>substrate-Bergman +<br>substrate-Pochhammer<br>FK Ch. XII §VI<br>multi-week derivation |

| Candidate                                                | Form                                                                                                       | Cal #29 audit                                                                                     | Cal #35 audit                                                                                                           |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Composite v0.4 Term 2<br>( $N_c^4 / 2^N N_c$ )           | substrate-Casey-#14-codim-4 +<br>substrate-color                                                           | substantive PARTIAL FORWARD (factor identification per Casey #14 STANDING + Casey #13 cross-link) | substantive PARTIAL substrate-mechanism FORCING candidate; multi-week explicit Casey #13 + Casey #14 cascade derivation |
| Composite v0.4 sum = 207 at 0.112%                       | composite $n_C \cdot (5/2)_3 + N_c^{4/2} N_c$                                                              | substantive PARTIAL FORWARD (component substantive substrate-mechanism candidates)                | substantive PARTIAL substrate-mechanism FORCING; multi-week joint Lyra + Keeper + Elie per Cal #189                     |
| IP-5 W mass ( $n_C \cdot m_p \cdot N_{\max} / 2^N N_c$ ) | substrate-genus + $m_p$ anchor +<br>substrate-fine-structure +<br>substrate-Cartan-base<br>substrate-color | substantive PARTIAL FORWARD (factor identification per Saturday May 28 + Casey #5 Integer Web)    | substantive PARTIAL substrate-mechanism FORCING; multi-week per Cal #189                                                |

### 3. RIGOROUS-Tier Promotion Path per Cal #29

Per Cal #29 + Cal #35 STANDING + K3 v0.16 Gate 9 RIGOROUS closure path:

Forward-derivation requirement: substrate-mechanism candidate FORM must be derivable from substrate first-principles WITHOUT reference to observed value.

RIGOROUS-tier candidate substrate-mechanism candidates: 1. Composite v0.4 Term 1 substrate-Bergman + substrate-Pochhammer Schur II forward-derivable per FK Ch. XII §VI multi-week 2. SSG-8 Mersenne ladder substrate-primary cascade rank  $\rightarrow N_c \rightarrow g$  forward-derivable per substrate-Mersenne map + K207 PASS

PARTIAL FORWARD substrate-mechanism candidate: Composite v0.4 Term 2 substrate-deficit per Casey #13 + Casey #14 STANDING substrate-codim-4 — substantive factor identification per Casey-named principles cross-link; multi-week explicit FORCING derivation per Cal #189.

Mode 5 form-fitting REJECTED candidates (Cal #35 STANDING brakes): - Cat 6 substrate-symplectic 4 substrate-canonical substrate-invariants at  $V_{\text{fund}}$  (all = unity) - Cat 5 substrate-multipole  $1/(2k+1)$  integer-INDEX coincidence

### 4. Substantive RIGOROUS Promotion Audit Status

Per K3 v0.16 Gate 9 RIGOROUS closure substantive promotion criteria:

| Candidate             | Cal #29 forward-derivation                                     | Cal #35 mechanism-forcing                         | RIGOROUS status                                  |
|-----------------------|----------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|
| Composite v0.4 Term 1 | YES (FK Ch. XII §VI substrate-Bergman + Schur II Pochhammer)   | substantive substrate-mechanism FORCING candidate | FRAMEWORK $\rightarrow$ RIGOROUS multi-week path |
| Composite v0.4 Term 2 | PARTIAL (Casey #13 + Casey #14 STANDING factor identification) | PARTIAL substrate-mechanism FORCING candidate     | FRAMEWORK candidate; multi-week per Cal #189     |
| SSG-8 Mersenne        | YES (substrate-Mersenne map + K207 PASS A-tier)                | substantive substrate-mechanism FORCING candidate | A-tier $\rightarrow$ RIGOROUS multi-week path    |

| Candidate                       | Cal #29<br>forward-derivation                                       | Cal #35<br>mechanism-forcing                  | RIGOROUS status                              |
|---------------------------------|---------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|
| IP-5 W mass                     | PARTIAL (factor identification per substrate-natural decomposition) | PARTIAL substrate-mechanism FORCING candidate | FRAMEWORK candidate; multi-week per Cal #189 |
| Cat 6<br>substrate-symplectic 4 | NO (walked back)                                                    | NO (substrate-canonical unity = NOT 1/g)      | REJECTED Cal #35<br>STANDING                 |
| Cat 5 substrate-multipole       | NO (integer-INDEX coincidence)                                      | NO (form-fitting per Cal #35)                 | REJECTED Cal #35<br>STANDING                 |

## 5. Closure v0.1

L6 K3 Gate 9 Cal #29 question-shape audit RIGOROUS v0.1: substantive Thursday afternoon substrate-mechanism candidates audit per Cal #29 + Cal #35 STANDING discipline.

substantive RIGOROUS-tier promotion candidates: Composite v0.4 Term 1 (substrate-Bergman + substrate-Pochhammer) + SSG-8 Mersenne ladder (K207 PASS A-tier).

substantive PARTIAL FORWARD candidates: Composite v0.4 Term 2 (substrate-deficit per Casey #13) + IP-5 W mass.

substantive Cal #35 STANDING REJECTED candidates: Cat 6 substrate-symplectic 4 substrate-canonical + Cat 5 substrate-multipole integer-INDEX coincidence.

Tier: L6 K3 Gate 9 Cal #29 question-shape audit RIGOROUS v0.1 substantive promotion candidates identified per Cal #29 + Cal #35 STANDING discipline; multi-week explicit FORCING derivation per Cal #189.

— Lyra, Thu 2026-06-04 11:18 EDT. L6 K3 Gate 9 Cal #29 question-shape audit RIGOROUS v0.1: substantive Thursday afternoon substrate-mechanism candidates audit; RIGOROUS-tier promotion candidates Composite v0.4 Term 1 + SSG-8 Mersenne; PARTIAL FORWARD Composite v0.4 Term 2 + IP-5 W mass; REJECTED Cat 6 substrate-symplectic + Cat 5 substrate-multipole per Cal #35 STANDING; multi-week per Cal #189.